Assignment 1: Opinion Network Formation

Team Name: Complex Network of Dynamic People

September 17, 2026

GitHub Link to Code

[TODO: Insert GitHub repository link here. Repo should contain build_network.py, the outputs/ and figures/ folders, and any additional scripts from teammates (item-network, null-model validation).]

1 Dataset Documentation

The data cleaning and encoding pipeline described below was implemented as part of the network construction workstream (see `build_network.py`); this section documents and elaborates on that process.

[TODO: KRIS: write up the description below (approx. 1 page), and add your own new material – descriptive statistics, distribution plots, a note on survey bias, etc. See outputs/encoded_responses.csv for the cleaned data.]

- Describe the raw survey: N respondents, 60 Likert-scale items across 4 topics (Technology, Education, Ethics/Society, Environment), 5-point scale (Strongly Disagree – Strongly Agree).
- Explain missing-data handling: "No Comments" entries were treated as missing responses rather than a neutral opinion. Respondents who left more than 50% of the survey blank were dropped (9 of 96 respondents removed, leaving 87). Remaining sparse gaps were filled with each item's class-average response.
- Explain the numeric encoding: Strongly Disagree = -2 ... Strongly Agree = +2.

[TODO: KRIS: add your own contribution here – descriptive statistics per topic (mean/spread), distribution plots, a discussion of self-report survey bias (social desirability, etc.), or other data exploration.]

2 Pipeline Followed

This section describes how the raw survey responses were transformed into an opinion network.

2.1 Data Cleaning and Encoding

Respondents who left more than 50% of the survey unanswered were removed (9 of 96 respondents), leaving 87 usable respondents. The string "No Comments" was treated as a missing value rather than a neutral opinion. Likert responses were mapped to an ordinal numeric scale from -2 (Strongly

Disagree) to +2 (Strongly Agree), and any remaining sparse gaps were filled with that item's class-average response.

2.2 Similarity Measure

Pairwise opinion similarity between respondents was computed using the **Pearson correlation coefficient** across their answer vectors, rather than cosine similarity or Euclidean distance. Correlation was chosen because it captures whether two respondents' opinions rise and fall *together* across statements, and is invariant to a respondent's individual tendency to answer more or less extremely overall – a scale bias rather than a genuine opinion difference.

2.3 Graph Construction

The dense 87×87 similarity matrix was sparsified using **mutual k-Nearest-Neighbours** with $k = 8$: an edge between respondents i and j was retained only if each was among the other's top-8 most similar peers. This produces a graph reflecting genuinely reciprocal opinion closeness, avoids an uninformative fully-connected graph, and sidesteps the need to hand-tune a similarity threshold.

In addition to the overall network (built from all 60 items), four topic-specific sub-networks were constructed using the same procedure restricted to each topic's 15 items, to allow cross-topic comparison of clustering structure.

2.4 Community Detection

Louvain modularity maximization was applied to the overall network to identify opinion communities. The same procedure was applied independently to each topic sub-network for comparison.

[TODO: PRATHISH (Statistical Validation): briefly describe your null-model validation methodology here (1 short paragraph) – e.g. "To assess whether the observed community structure is statistically meaningful, modularity was compared against a null distribution generated from N random graphs matched on node count, edge count, and degree sequence..."]

3 Analysis and Visualizations

3.1 Overall Network Structure

3.2 Topic Sub-Networks and Cross-Topic Comparison

Table **??** summarizes the four topic-specific sub-networks.

3.3 Opinion Bridges

To identify respondents who structurally connect otherwise separate opinion communities, betweenness centrality was computed for every node, and the community diversity of each top-scoring node's immediate neighborhood was examined (Table **??**).

3.4 Community Opinion Archetypes

For every community with at least 4 members, the average deviation from the class-wide opinion score was computed per topic, producing interpretable "opinion archetypes" for each cluster (Figure **??**).

Metric	Value
Nodes	87
Edges	134
Density	0.036
Average clustering coefficient	0.157
Average path length (largest component)	3.86
Diameter (largest component)	9
Number of connected components	18
Largest component size	66
Number of Louvain communities	25
Modularity	0.578

Table 1: Overall opinion network metrics.

Topic	Edges	Density
Technology	201	0.054
Education	176	0.047
Ethics/Society	227	0.061
Environment	193	0.052

Table 2: Topic sub-network metrics (87 nodes each).

[TODO: KRIS: insert item-network figures and metrics here (which survey questions cluster together).]

[TODO: PRATHISH (Statistical Validation): insert null-model validation figures here (e.g. histogram of null-model modularity scores with observed modularity marked, significance test results).]

4 Results and Discussion

4.1 Person-Opinion Network (this workstream)

The overall opinion network fragments into approximately 25 Louvain communities at a modularity of 0.578, indicating that class opinion is organized into many small, fine-grained groups rather than 2–3 broad ideological camps.

Comparing topic-specific community structure to the overall network via Adjusted Rand Index shows that **Education** (ARI = 0.181) and **Technology** (ARI = 0.101) opinions are the strongest predictors of a respondent's overall opinion-community membership, while **Ethics/Society** (ARI = 0.023) and **Environment** (ARI = 0.030) opinions are essentially uncorrelated with overall clustering. This suggests that ethics and environmental questions attract broad, cross-cutting agreement or disagreement that does not track the same social groupings visible in tech/education opinions.

The bridge analysis further shows that the network's cohesion depends on a small number of specific individuals rather than broad consensus: respondent 81's immediate neighbors span 5 distinct communities, more than any other node, making this respondent (and the four others in Table **??**) structurally critical to keeping otherwise separate opinion groups connected.

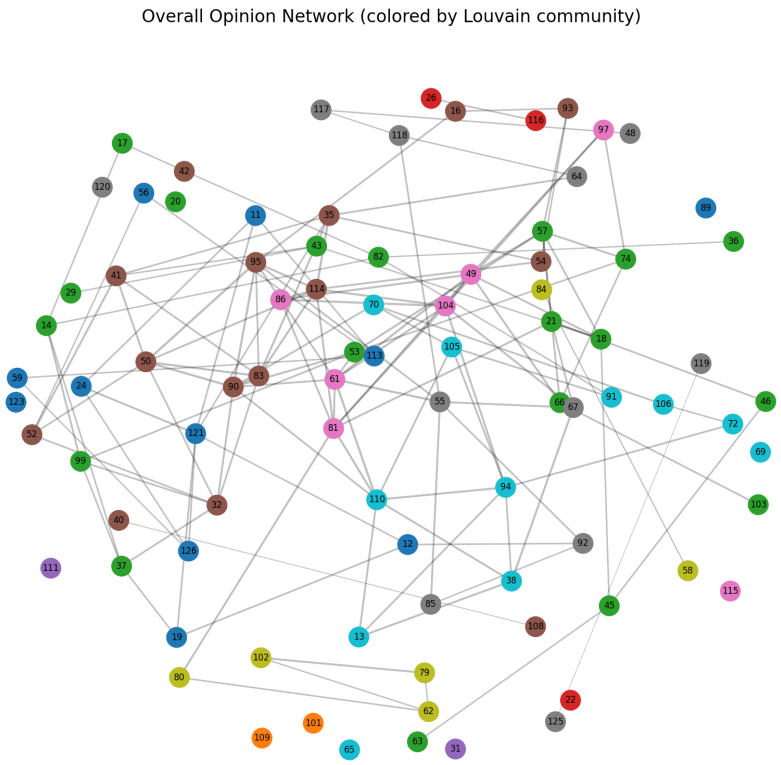


Figure 1: Overall opinion network, nodes colored by Louvain community.

The community opinion archetypes reveal clearly interpretable clusters: for example, community 0 (9 members) scores well below the class average on both Ethics/Society and Environment items – a coherent "skeptic" archetype – while community 16 (6 members) scores above average on every topic, representing a small "strong-consensus" cluster.

[TODO: KRIS: discuss item-network findings here – which opinion dimensions/clusters of questions emerged, and how they relate to the person-network findings above.]

[TODO: PRATHISH (Statistical Validation): discuss null-model validation findings here – is the observed community structure statistically significant compared to random graphs of the same size/density? Does this justify $k = 8$ as a sparsification parameter?]

4.2 Overall Synthesis

[TODO: ALL: once each subsection above is written, add 1 short paragraph tying the three analyses together – what does the combination of the person-network, item-network, and statistical validation tell us about how this class thinks, collectively?]

5 Individual Contribution

- **Shreem Shukla**: Cleaned and pre-processed the raw survey data (missing-value handling, dropping incomplete responses, Likert-to-numeric encoding); built the full person-opinion

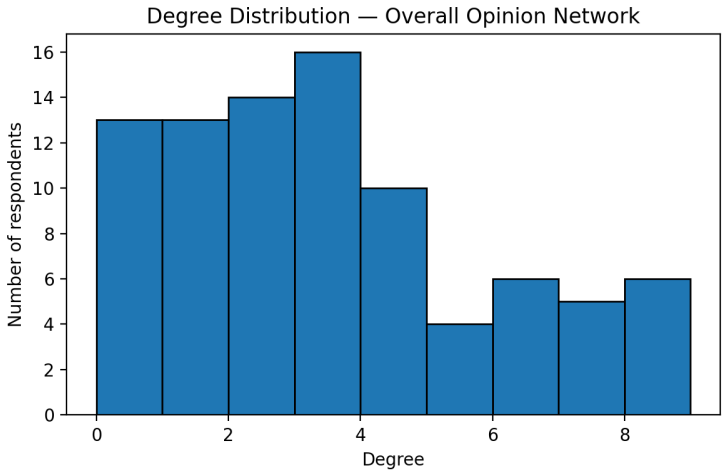


Figure 2: Degree distribution of the overall opinion network.

Respondent	Betweenness Centrality	Own Community	Neighbor Communities Spanned
81	0.083	16	5
35	0.090	12	3
90	0.062	12	4
55	0.074	18	3
121	0.083	0	2

Table 3: Top opinion-bridge respondents by betweenness centrality.

network construction pipeline (Pearson similarity, mutual k-NN graph construction, Louvain community detection, degree/betweenness/ eigenvector centrality); performed the cross-topic ARI comparison, the opinion-bridge analysis, and the community opinion-archetype analysis. Wrote the Pipeline Followed section, the data-cleaning portion of Dataset Documentation, and this report skeleton.

- **Kris Jain**: **[TODO: fill in – e.g. "Documented and described the dataset (built on the cleaning pipeline above), added descriptive statistics and distribution analysis; built the item-item opinion network identifying clusters of related survey questions; wrote the remainder of Dataset Documentation."]**
- **Prathish Ghosh**: **[TODO: fill in – e.g. "Designed and ran null-model statistical validation of network community structure; performed sensitivity analysis across k values; contributed to Results and Discussion."]**

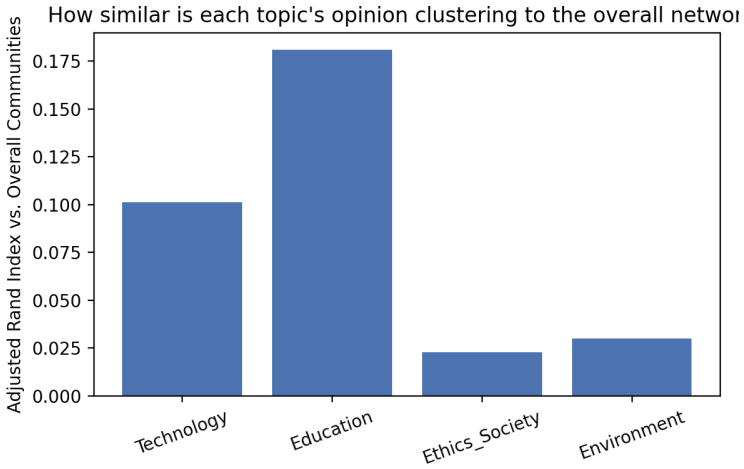


Figure 3: Adjusted Rand Index between each topic's community structure and the overall network's community structure.



Figure 4: Opinion bridges (red-ringed nodes) overlaid on the overall network.

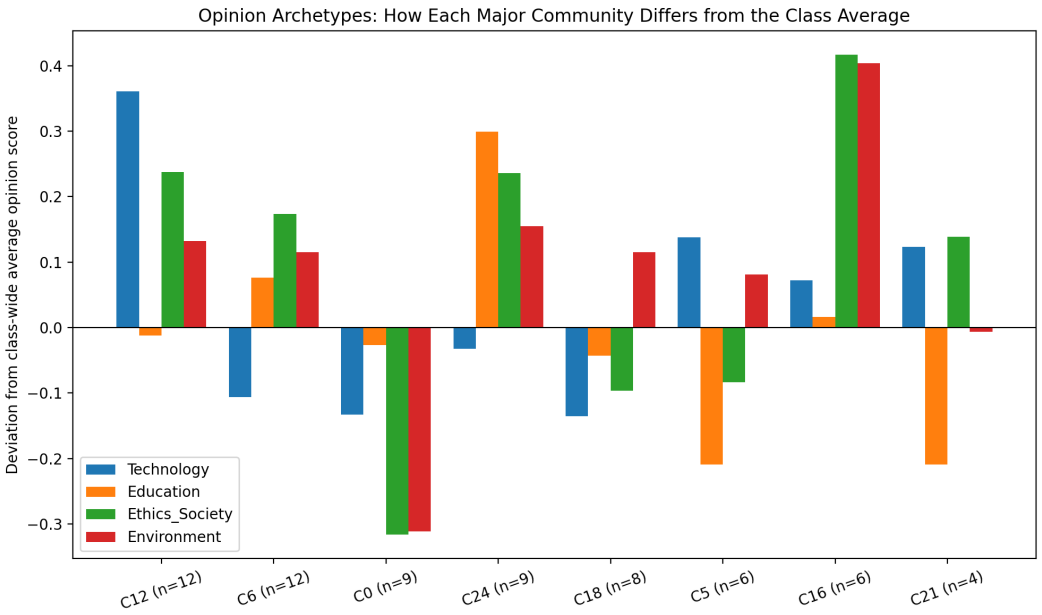


Figure 5: Opinion archetypes: deviation of each major community from the class-wide average, by topic.